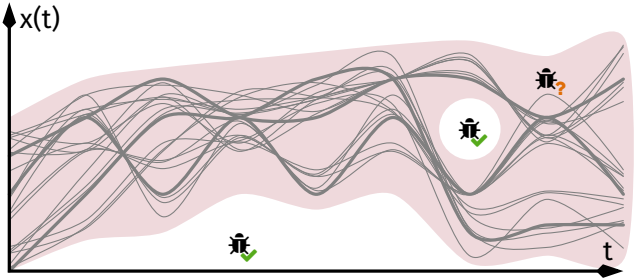

Abstract Interpretation

Software Quality Assurance — Static Code Analysis, II | Florian Sihler | December 11, 2024

Software Engineering
Programming Languages

universität ulm

1. The Why

The Why

```
public static void main(String[] args) {
   int a = 1;
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   if (r > 5) {
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- We want to proof, that a program satisfies certain properties

Origins

Static Analysis

Based on the amazing "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" by Miné [Min17], https://www.di.ens.fr/~cousot/AI/, and [Bal+18; GR22]

Origins

1949: First Checks
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Non-trivial Properties are undecidable [Ric53]

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1967 & 69: Logical Foundation
Floyd [Flo67], Hoare [Hoa69]
But: No Automation

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Deductive Methods

Model Checking

Symbolic Execution

Abstract Interpretation

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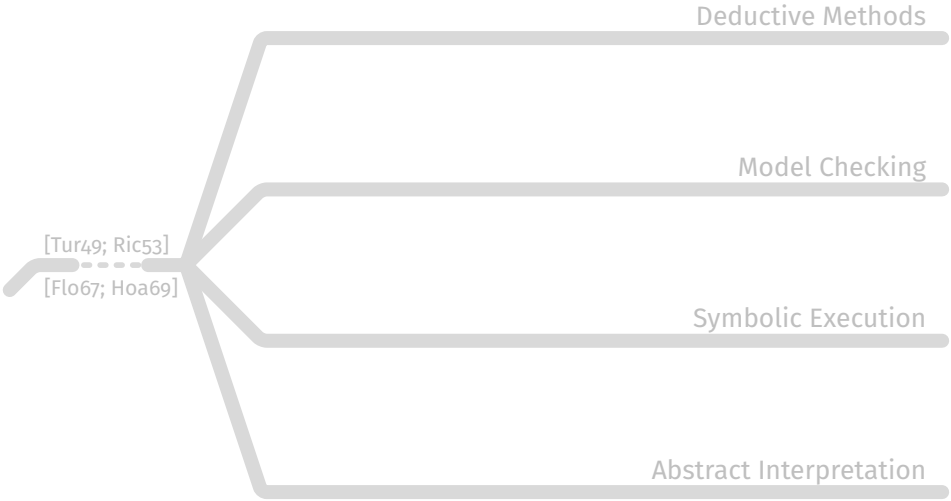

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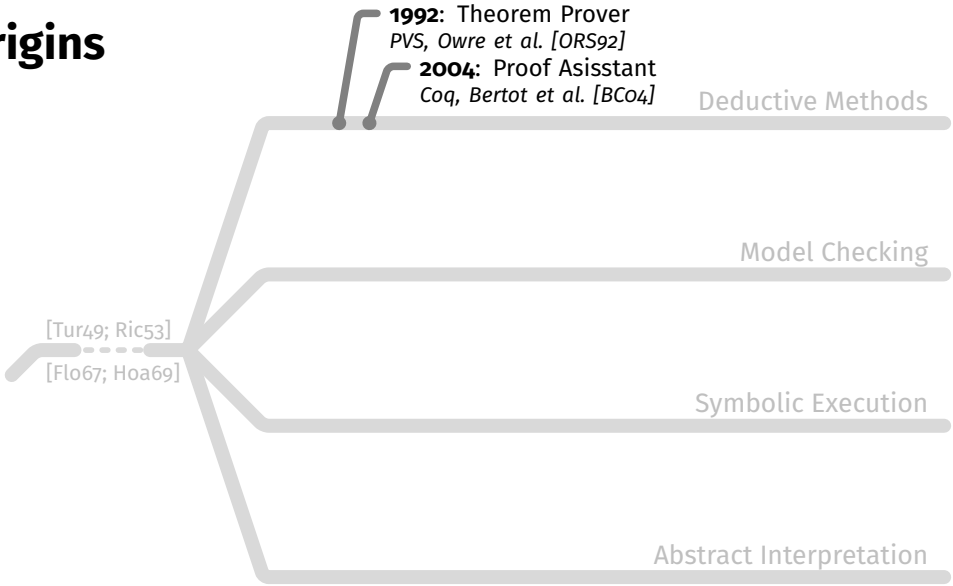

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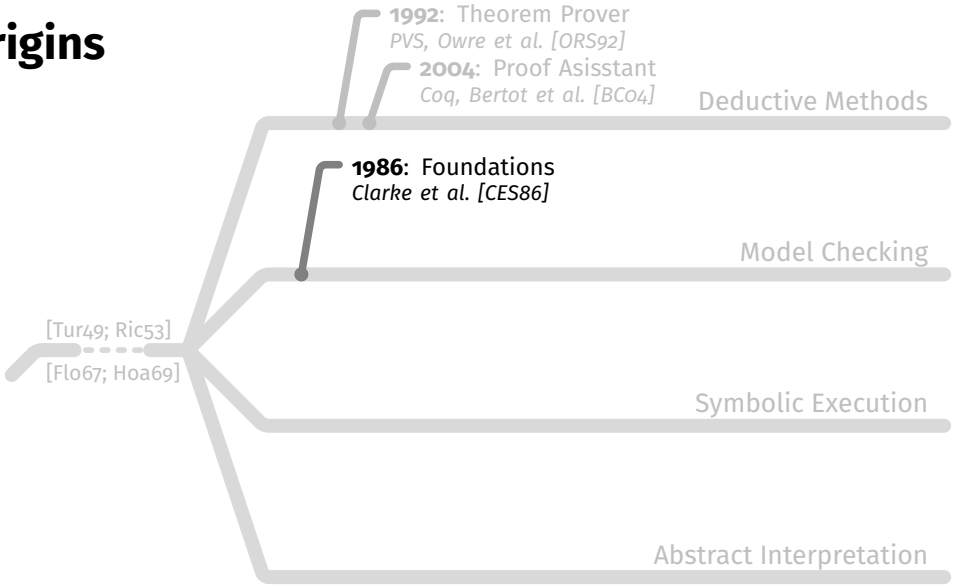

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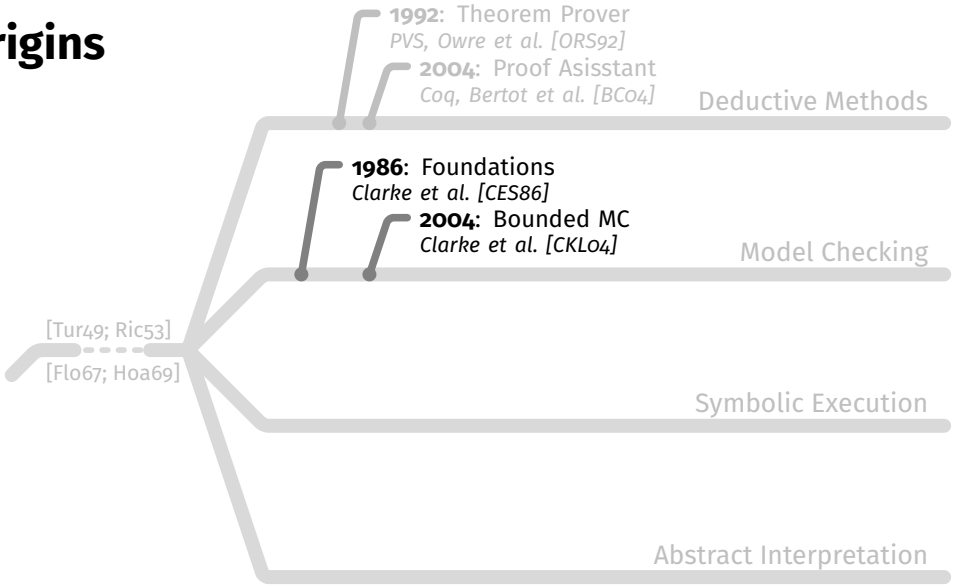

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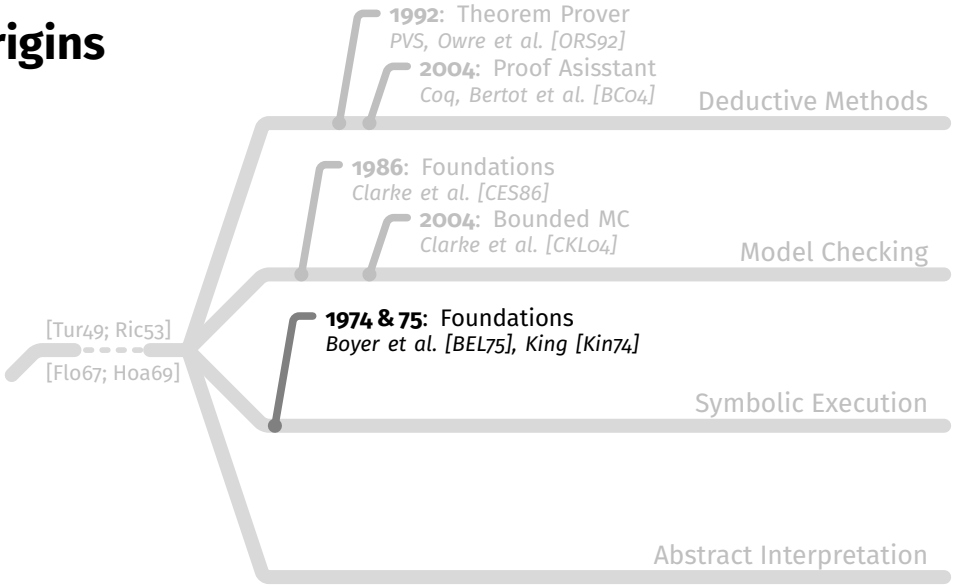

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2008: Automation
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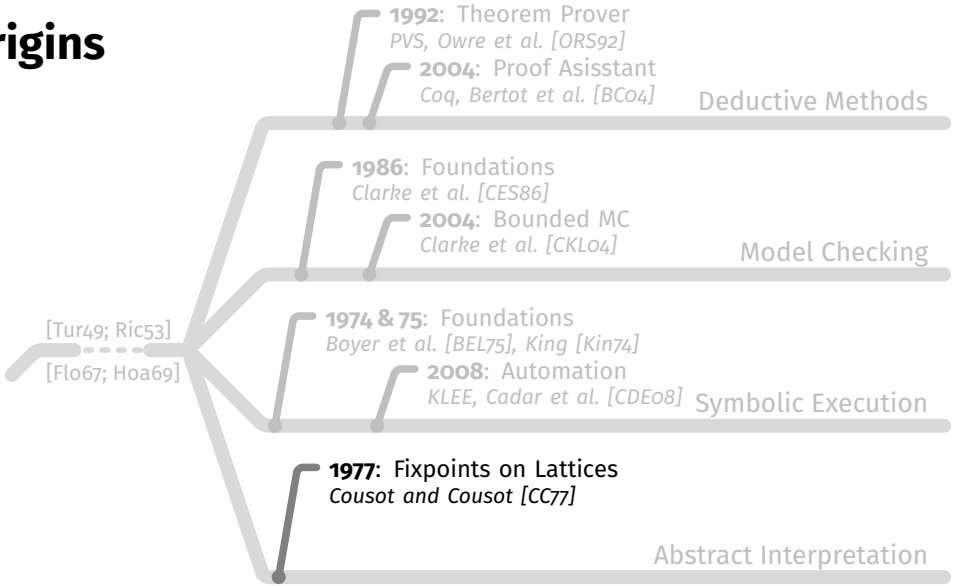

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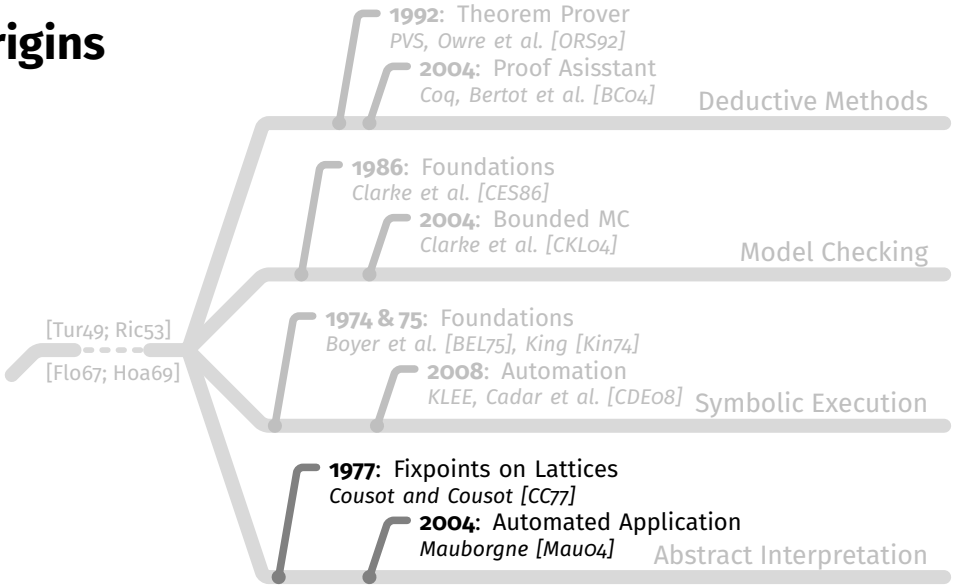

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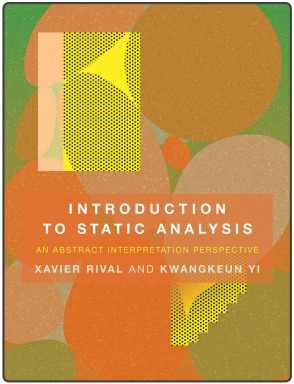

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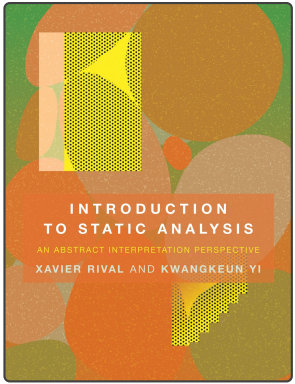
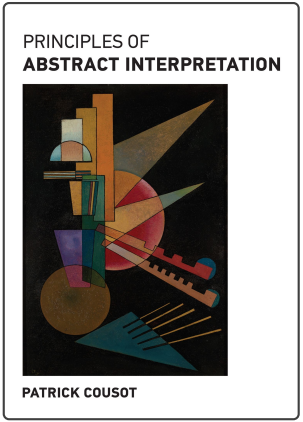


Formal Foundations [Cou21]



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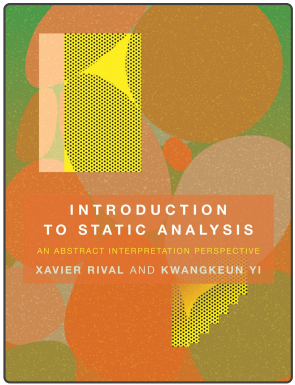

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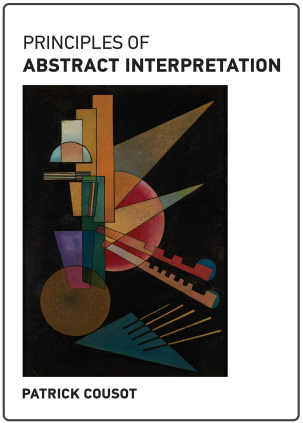
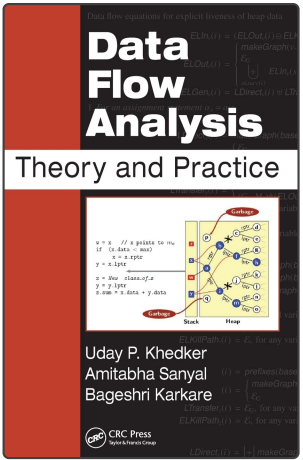


Dataflow Perspective [KSK09]



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2. The How

Abstract Interpretation

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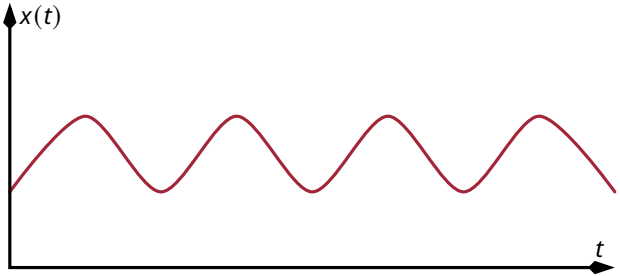

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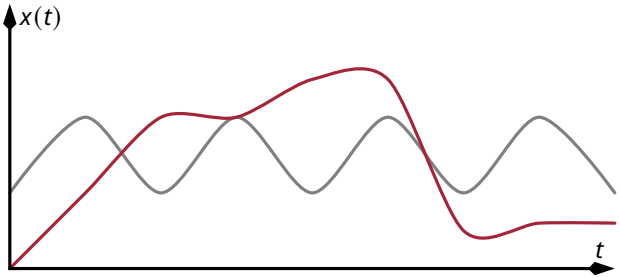

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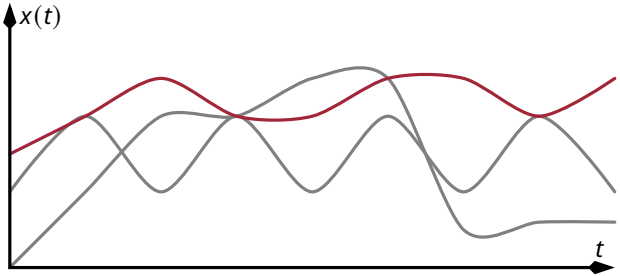

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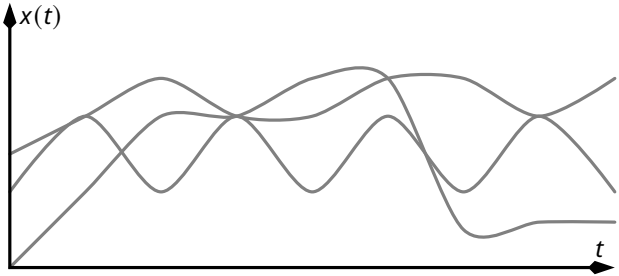

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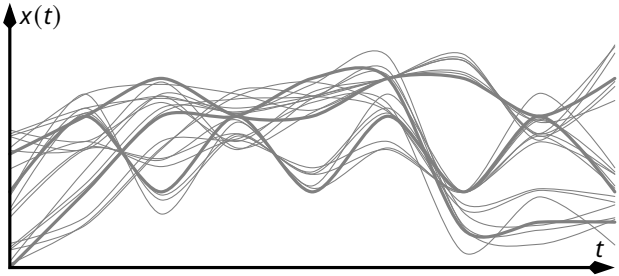

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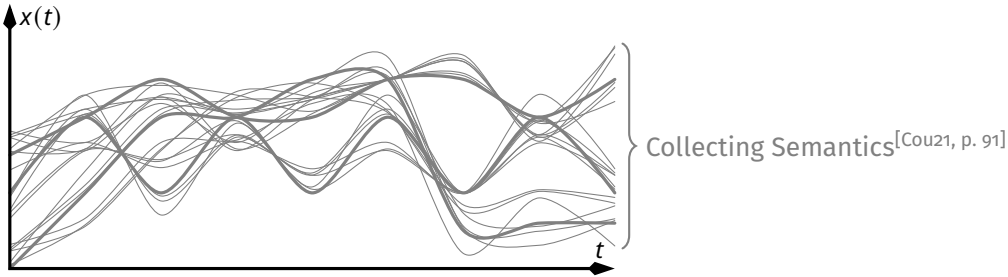

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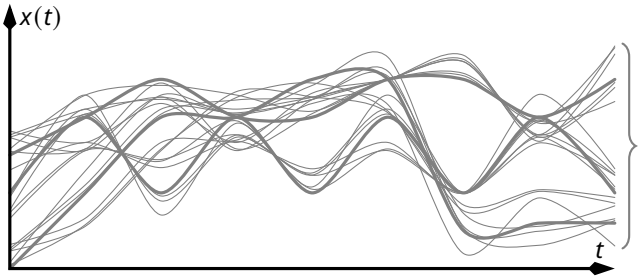

Collecting Semantics[Cou21, p. 91]

- Maybe impossible to compute statically
- … or very expensive (▸ *dynamic*)
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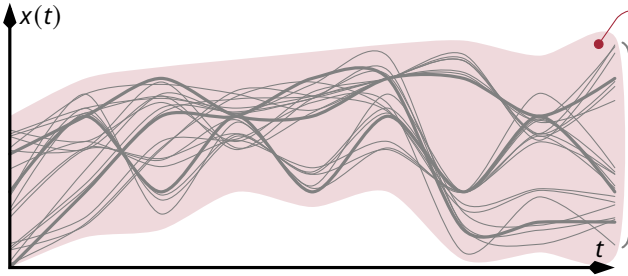

(Trace) Abstraction[Cou21, p. 92]
just one of many

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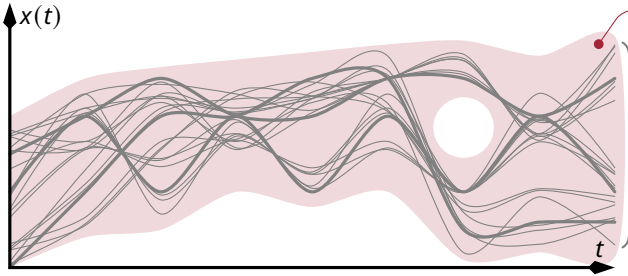

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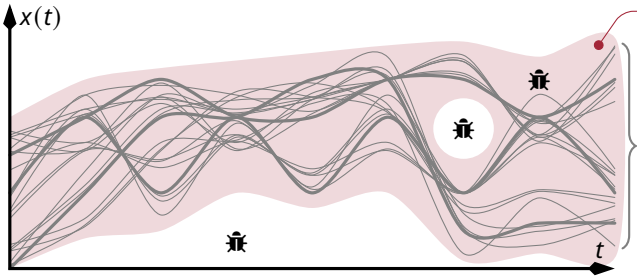

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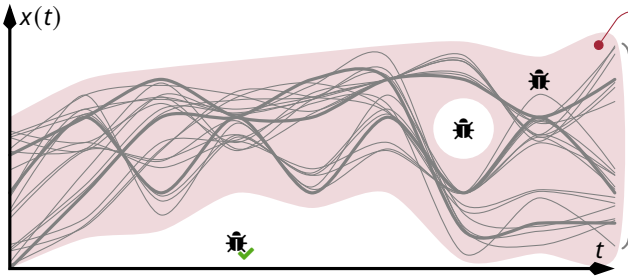

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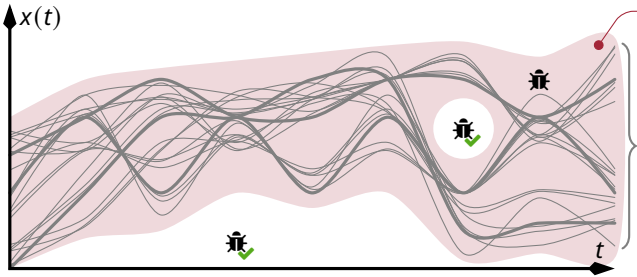
See "A casual introduction to Abstract Interpretation" [Cou12]

Abstract Interpretation



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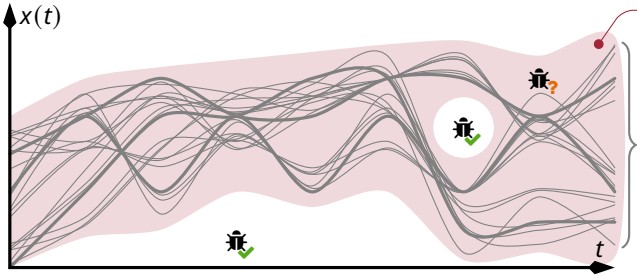

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Terminology

- **Property**

"Principles of Abstract Interpretation" [Cou21, p. 15],"Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 18]

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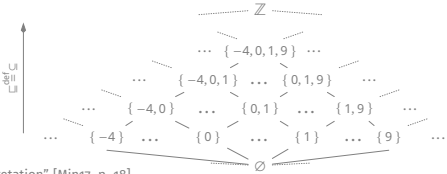

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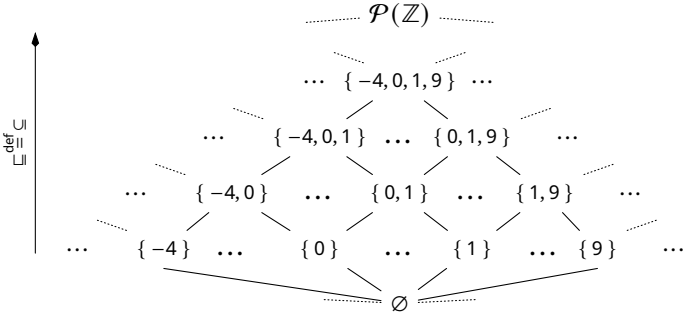
Chains and Lattices



"Lattice theory" [Bir67], see also sublattices [Min17, p. 25]

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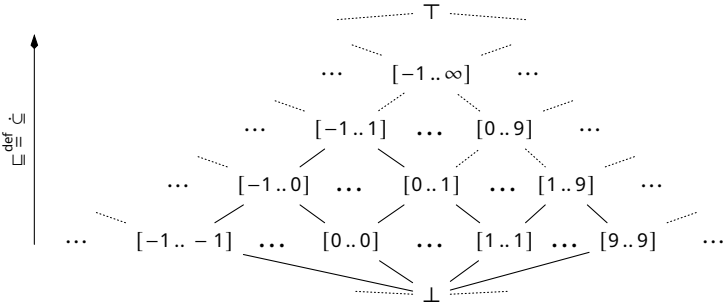

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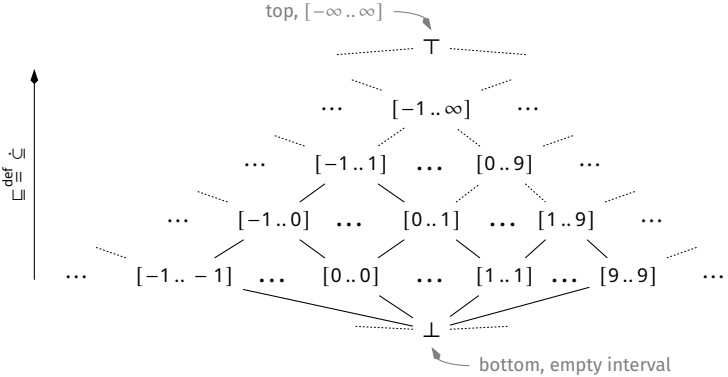

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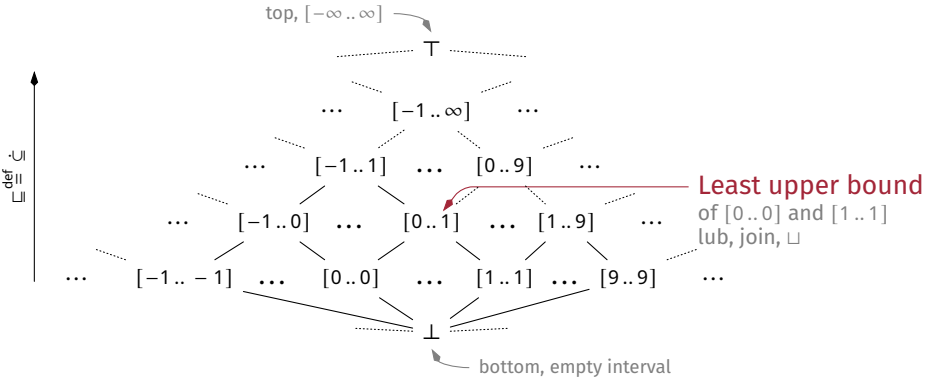

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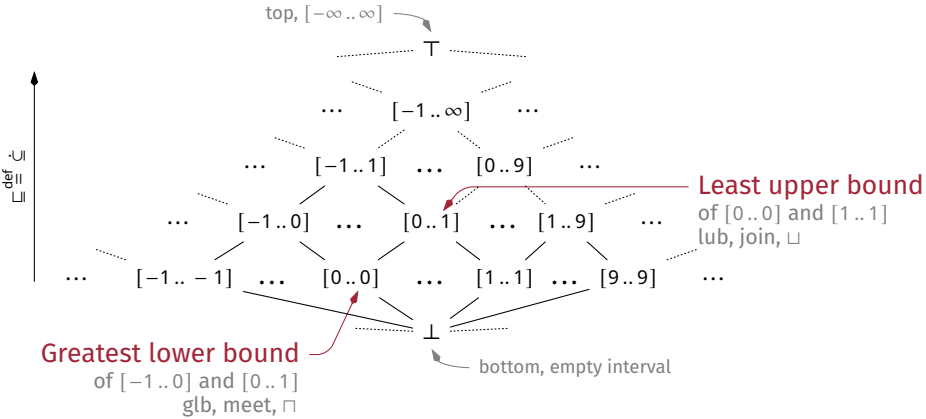

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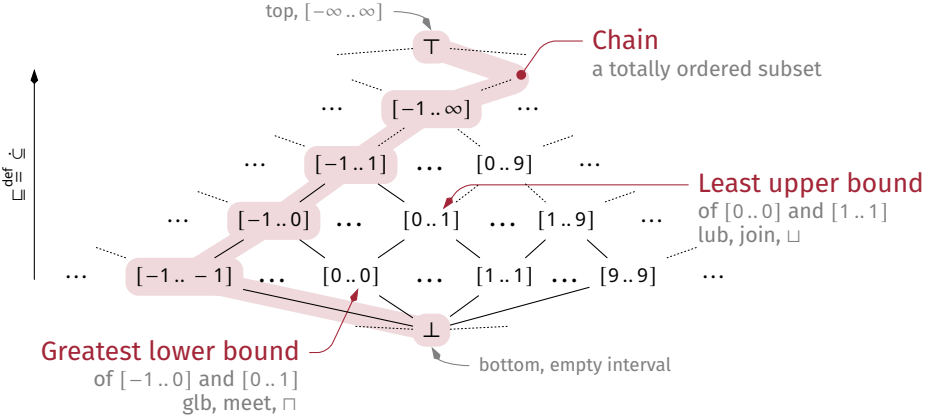

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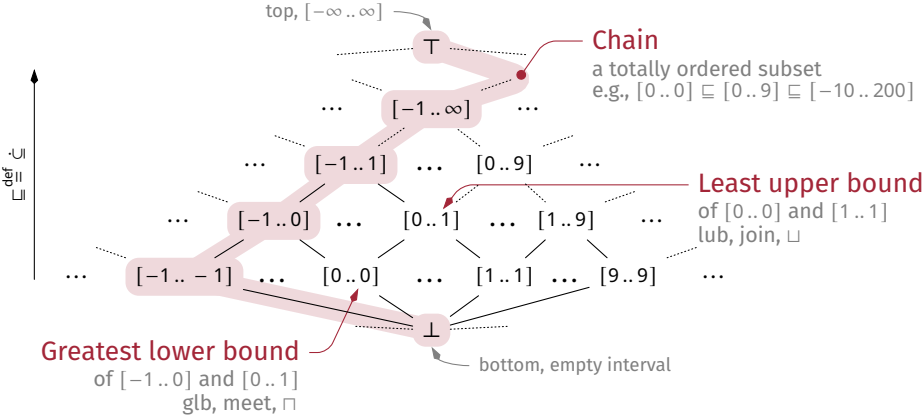

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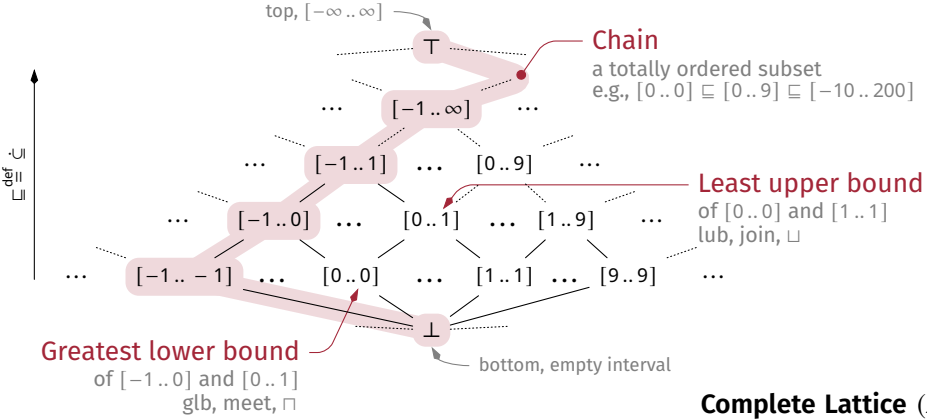

Complete Lattice $(X, \sqsubseteq, \sqcup, \sqcap, \bot, \top)$

"Lattice theory" [Bir67], see also sublattices [Min17, p. 25]

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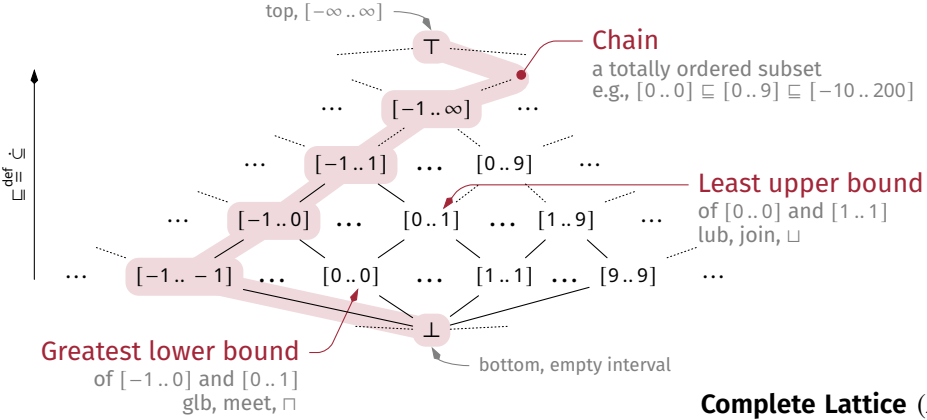

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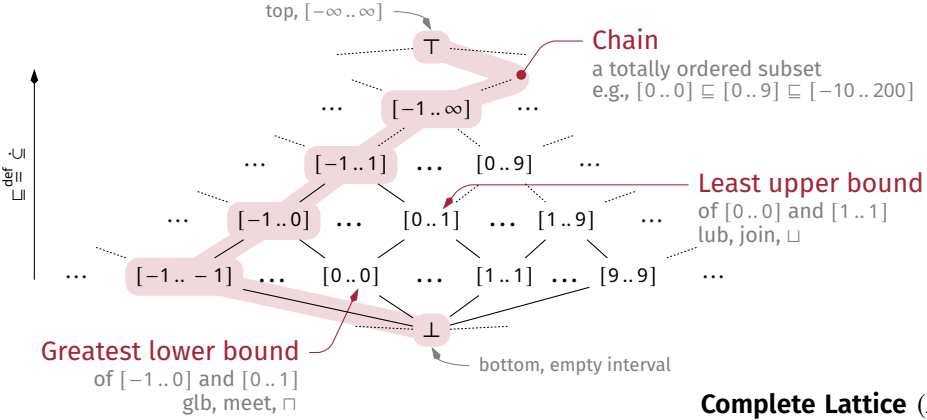

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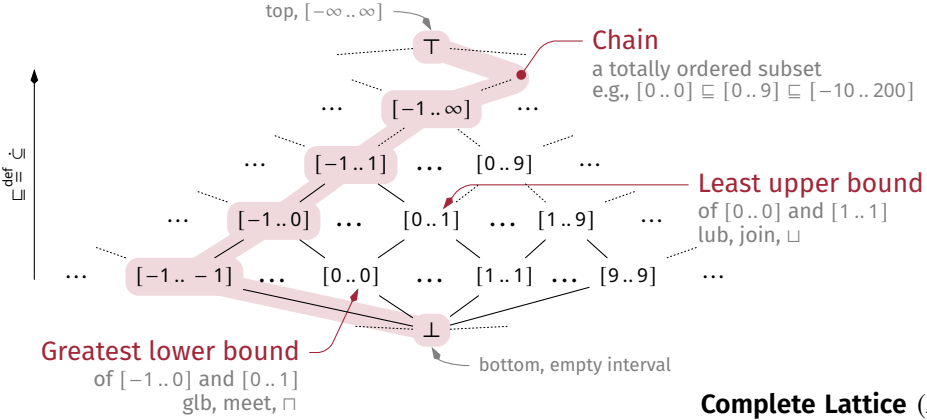

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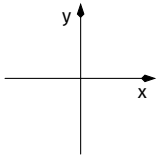
Abstract Domains Numerical



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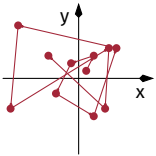

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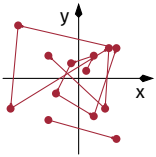

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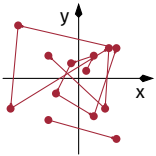

Collecting Semantics

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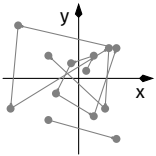

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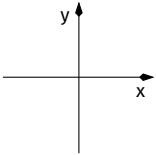

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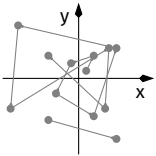

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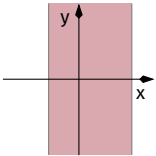

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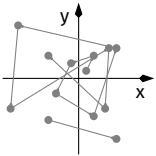

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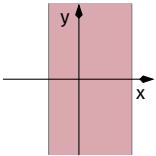

Intervals $x \in [a\,..\,b]$

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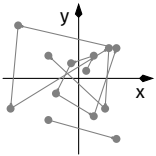

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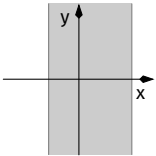

Intervals $x \in [a\,..\,b]$

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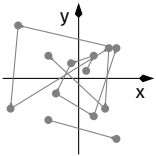
Abstract Domains Numerical



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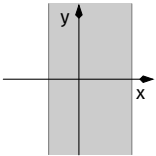

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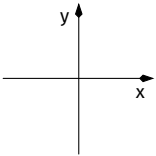

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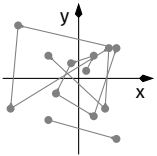

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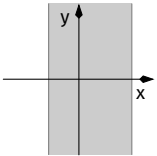

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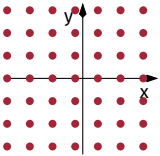

Simple Congruences

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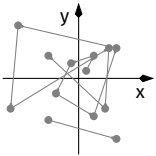

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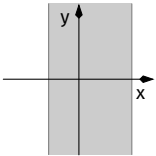

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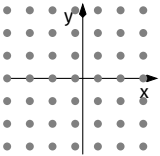

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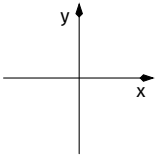

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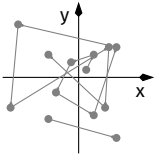

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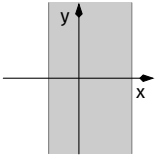

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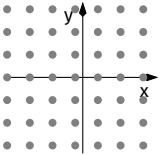

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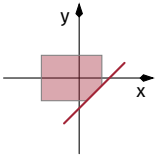

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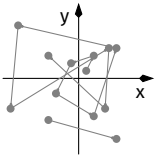

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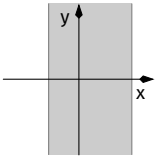

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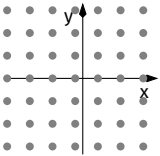

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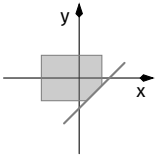

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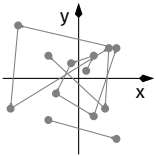

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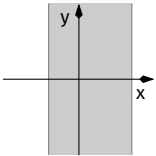

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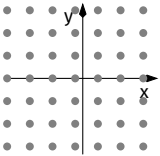

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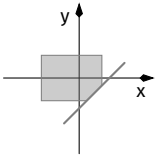

Pentagons

- Octagons

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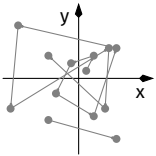

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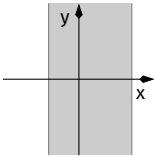

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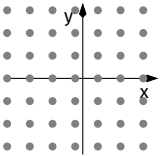

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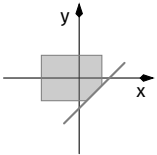

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- Ellipses

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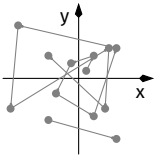

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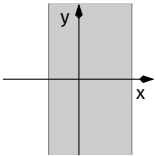

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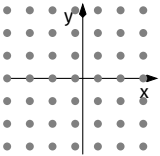

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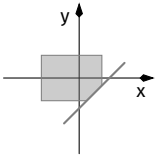

Pentagons

- Octagons
- Ellipses
- Exponentials

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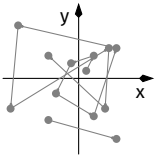

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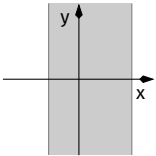

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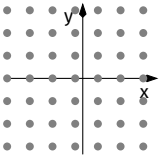

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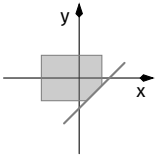

Pentagons

- Octagons
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- Signs

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Sign Analysis

Simple Sign Domain

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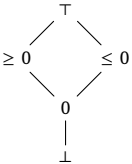

Sign Analysis

Simple Sign Domain

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int a = 0;
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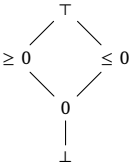

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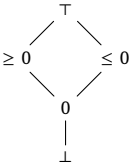

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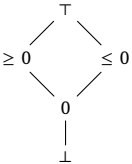

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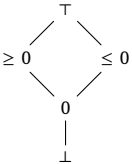
$\wr a = 0 \int$

$\wr b \geq 0 \int$

$\wr c \geq 0 \quad (= 0 \ + \ \geq 0) \int$



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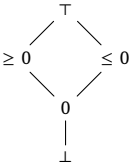

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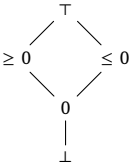
- But how to handle control flow? Loops? Recursion?



Fixpoints

"A lattice-theoretical fixpoint theorem and its applications." [Tar55], "Introduction to metamathematics" [Kle52], "Principles of Abstract Interpretation" [Cou21, p. 165]

Fixpoints

- For operators $f : X \to X$ a **fixpoint** is a $x \in X$ such that $f(x) = x$

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    reach a fixpoint, $f^p = f(f^p)$

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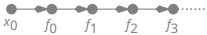
  - 

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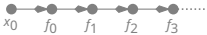
  - 

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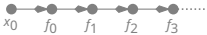
  - 

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  for complete, nonempty lattices
  Tarski's Theorem
- Analyzing, e.g. loops, we "go up" the lattice until we reach a least fixpoint

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Interval Analysis, I

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int x = 0;
while(x < 2) {
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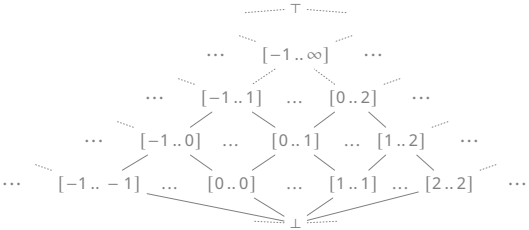

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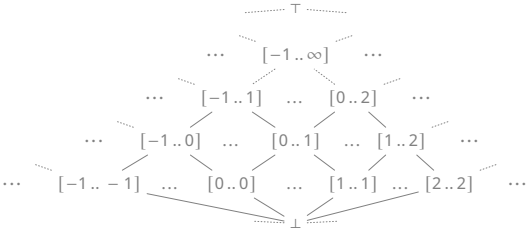

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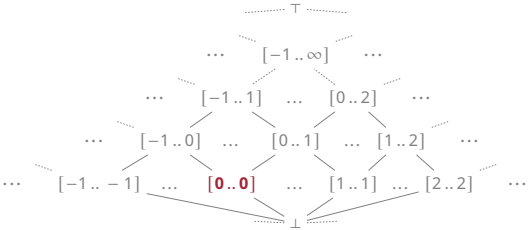

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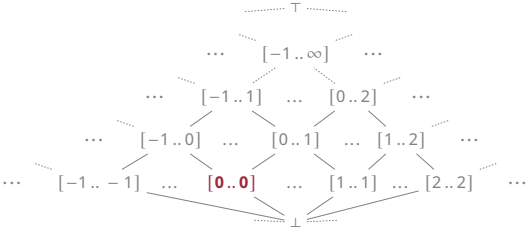

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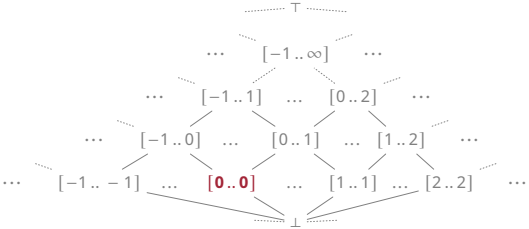

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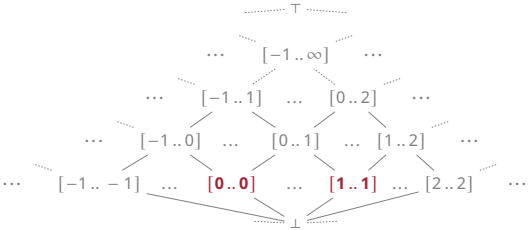

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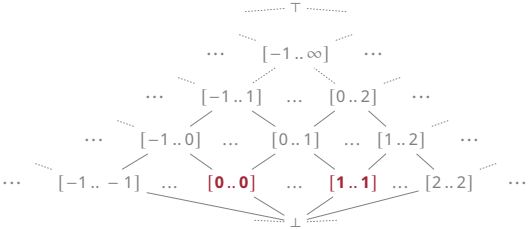

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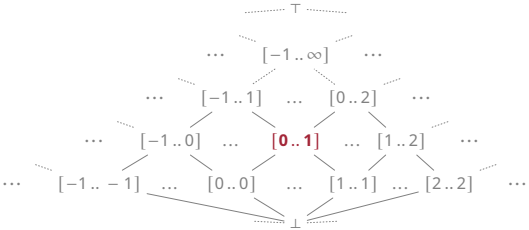

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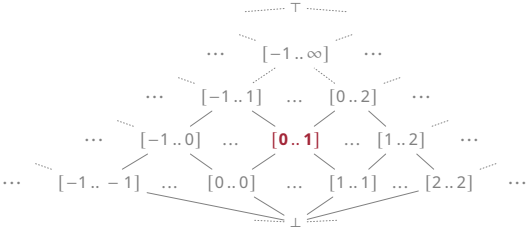

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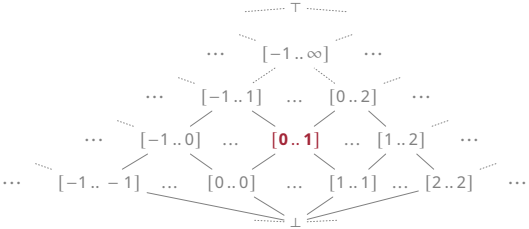

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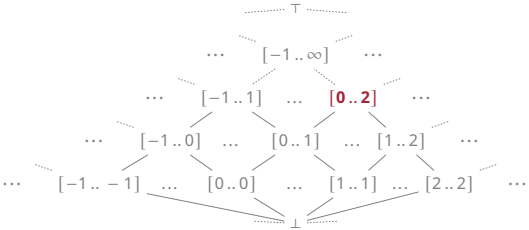

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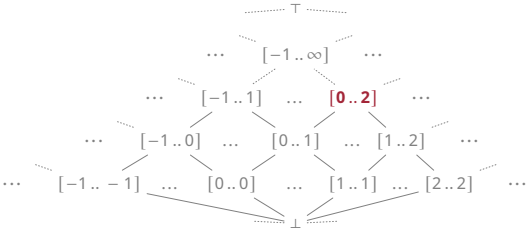

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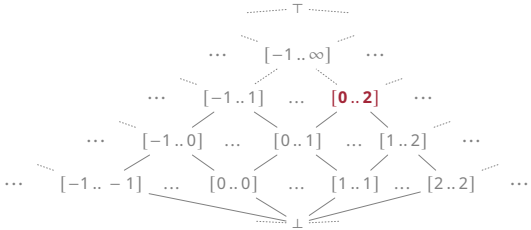

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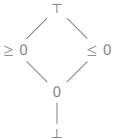

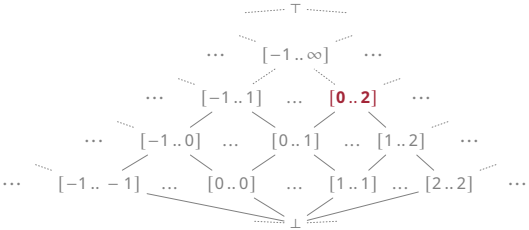

3. Semantics

Semantics

Shortened form of “Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation” [Min17, p. 46], craftinginterpreters.com/representing-code.html

Program Syntax (simplified)

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int x = 0;
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Numeric Constant $c \in \mathbb{I}$

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Loop

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Binary Expression

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Binary Expression

stm	::=	$V \leftarrow expr$	(assignment, $V \in \mathbb{V}$)
	|	$stm_1;\ stm_2$	(sequence)
	|	**while**$(cond)$ { stm }	(loop)
expr	::=	V	(variable, $V \in \mathbb{V}$)
	|	c	(constant, $c \in \mathbb{I}$)
	|	$expr_1 \diamond expr_2$	(bin. expr., $\diamond \in \{+, -, \ldots\}$)
cond	::=	b	(boolean, $b \in \mathbb{B}$)
	|	$expr_1 \bowtie expr_2$	(comparison, $\bowtie \in \{\leq, <, \ldots\}$)

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Atomic Expression Semantics

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$expr$	$::=$	V	(variable, $V \in \mathbb{V}$)
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- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

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- Additionally we can define evalCond(cond, envs) and evalStm(stm, envs)

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Denotational Semantics while loops

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while(*cond*) { *stm* }

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Suppose we start the loop with states *Start*

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iterate to find the least fixpoint [Min17, p. 52]

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Interval Analysis, II

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int x = 0;
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"Principles of Abstract Interpretation" [Cou21, p. 393], "Widening and narrowing operators for abstract interpretation" [CZ11]

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$\wr$ [pre] $x_1 \in [0\,..\,2] \quad ([0\,..\,1] \cup [1\,..\,2]) \,\wr \;\; \nabla \implies x_1 \in [0\,..\,\infty)$

$\wr$ [in] $x_2 \in [0\,..\,1] \quad ([0\,..\,1] \cap (-\infty\,..\,1]) \,\wr$

$\wr\, x_3 \in [1\,..\,2] \quad ([0\,..\,1] \oplus [1\,..\,1]) \,\wr$

$\wr$ [post] $x_4 \in [999999\,..\,\infty) \quad ([0\,..\,\infty) \cap [999999\,..\,\infty)) \,\wr$

- Fixpoint iteration can be *very* expensive, and may not stabilize
- *Widening* (∇) is crucial, computing an upper bound

"Principles of Abstract Interpretation" [Cou21, p. 393], "Widening and narrowing operators for abstract interpretation" [CZ11]

4. Soundness and Completeness

Rice's Theorem

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- We can not solve the halting problem
- We have to approximate the reality

The Confusion Matrix

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E.g., do we claim there is an error?

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E.g., is there really an error?

- **Precision:** $\mathbf{TP}/(\mathbf{TP} + \mathrm{FP})$ ("how many false alarms")
- **Recall:** $\mathbf{TP}/(\mathbf{TP} + \mathrm{FN})$ ("how many errors did we find")

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Abstract interpretation soundly over-approximates the program semantics

5. Outlook

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  combine abstract domains[Min17, p. 149]

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- Function calls
  require special handling[MJ12]
- Existing libraries allow for easy implementation
  LiSA[Fer+21], MOPSA[Jou+19], Apron[JM09]

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